Saliency Maps & Input-Space Interpretability

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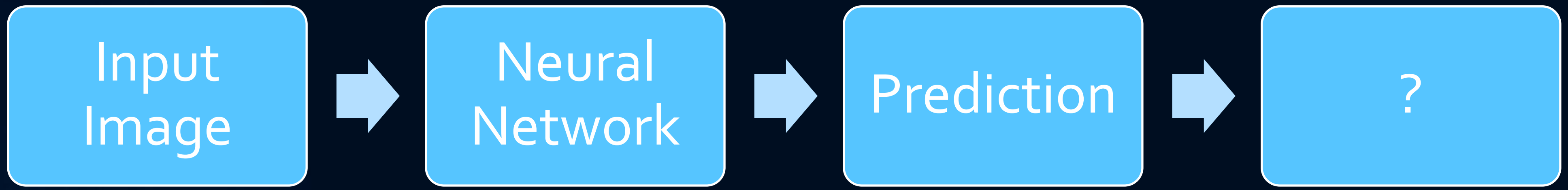
77304L

ARI5118 – DEEP LEARNING FOR COMPUTER VISION

WHY INTERPRETABILITY MATTERS

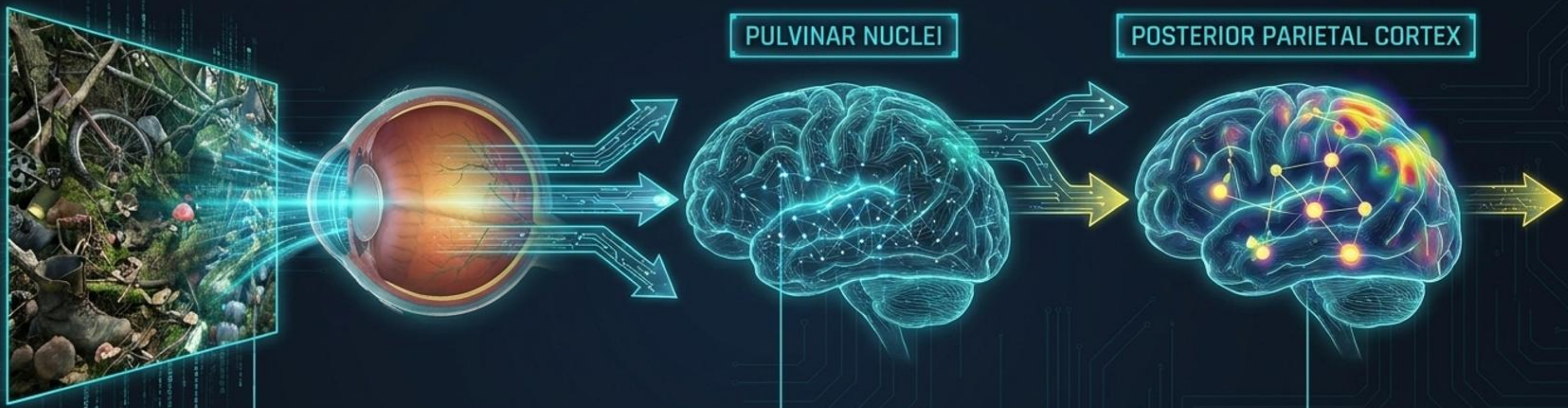


What Influenced this Decision?

High accuracy does <u>not</u> guarantee that reasoning is trustworthy.

THE BIOLOGICAL BLUEPRINT FOR MACHINE VISION

Primates interpret complex scenes in real-time despite slow neuronal hardware.
They achieve this via a serial strategy: a ‘focus of attention’ scans the scene.



PRE-ATTENTIVE

Bottom-up, sensory-driven, and task-independent processing.

TOPOGRAPHIC

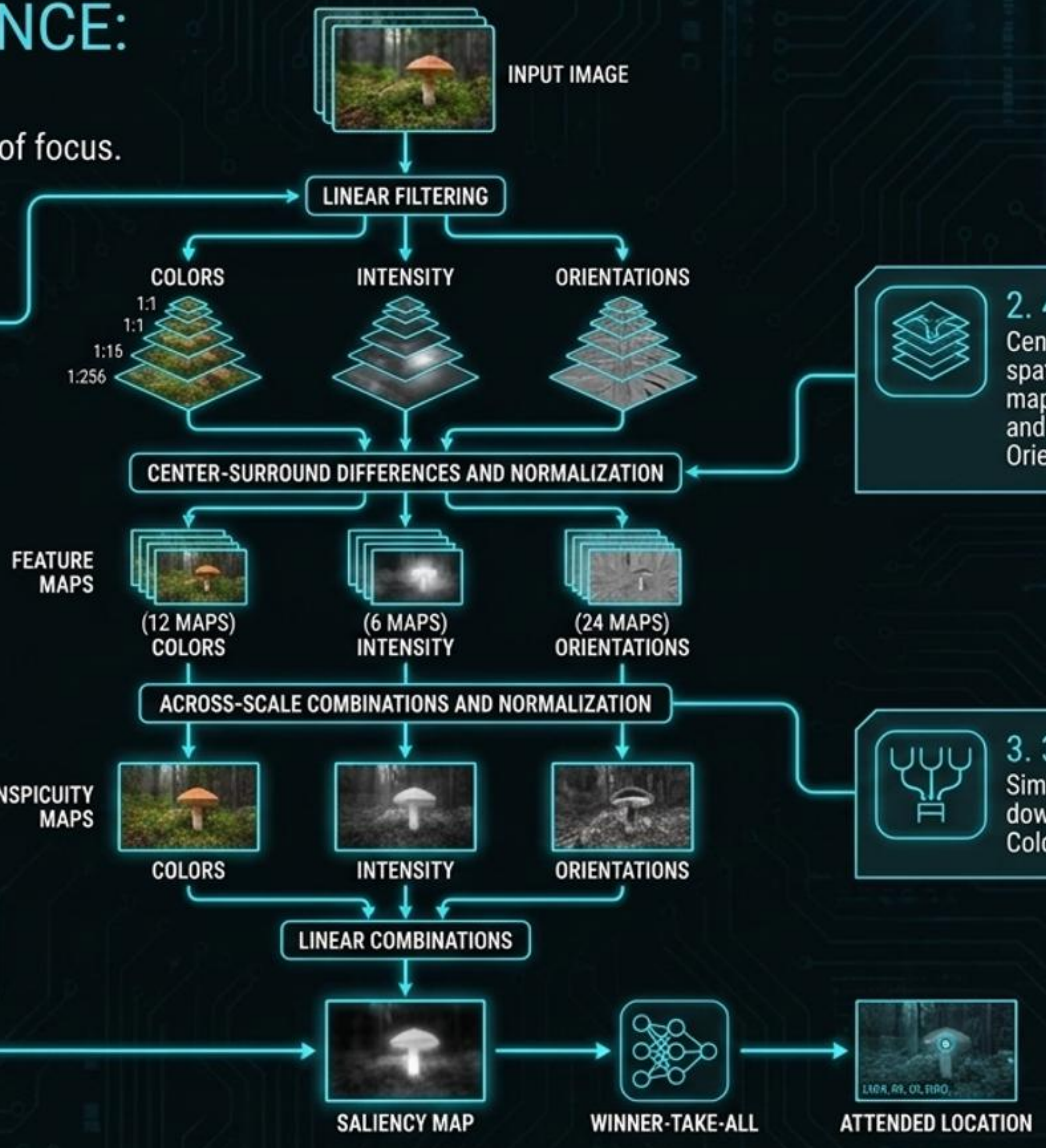
Spatial locations dynamically compete for overall conspicuity.

MASSIVELY PARALLEL

Rapid selection of a small number of interesting locations before passing data to slower, complex object-recognition stages.

CONSTRUCTING SALIENCE: THE 1998 PIPELINE

How raw pixels are filtered into a single point of focus.



1. LINEAR FILTERING

Input is analysed across 9 spatial scales (1:1 to 1:256 reduction)

2. 42 FEATURE MAPS

Centre-surround differences compute spatial contrast across Intensity (6 maps), Colour (12 maps using red/green and blue/yellow opponency), and Orientation (24 maps using Gabor filters)

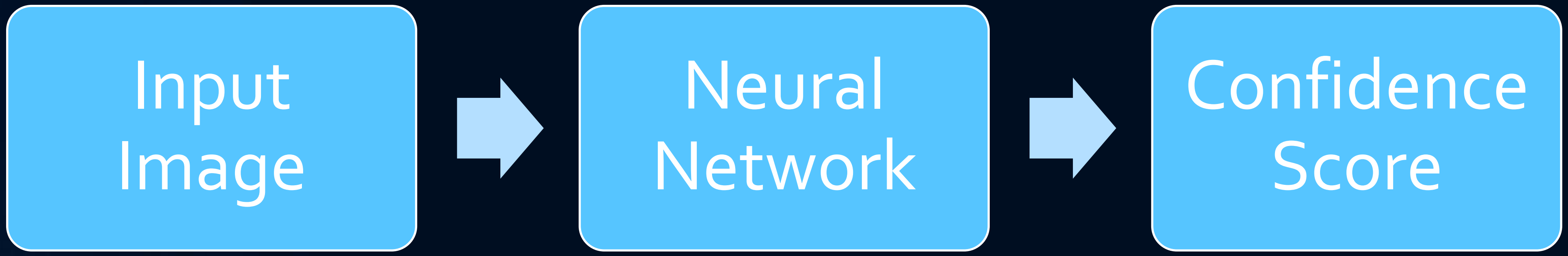
3. 3 CONSPICUITY MAPS

Similar features compete, normalising down to isolated channels for Intensity, Colour, and Orientation.

4. 1 SALIENCY MAP

Linear combination creates a single master map directing the winner-take-all network.

WHAT ARE INPUT GRADIENTS?



How does changing one pixel affect the final confidence score?

Large Change -> Important

Small Change -> Less Important

WHAT IS A SALIENCY MAP?

- A **saliency map** is the result of gathering all of the tiny "nudges" and displaying them as a single image.
- This map works in a similar way as a heat-sensing camera for the AI's brain.
- When the AI looks at pictures of animals, the saliency maps would highlight important areas of the animals such as ears and the beak of the flamingo.
- A saliency map can turn the mathematical importance of each pixel into a visual guide that humans can understand.

VANILLA SALIENCY MAPS

- Find the **derivative of the class score** with respect to *the input image*

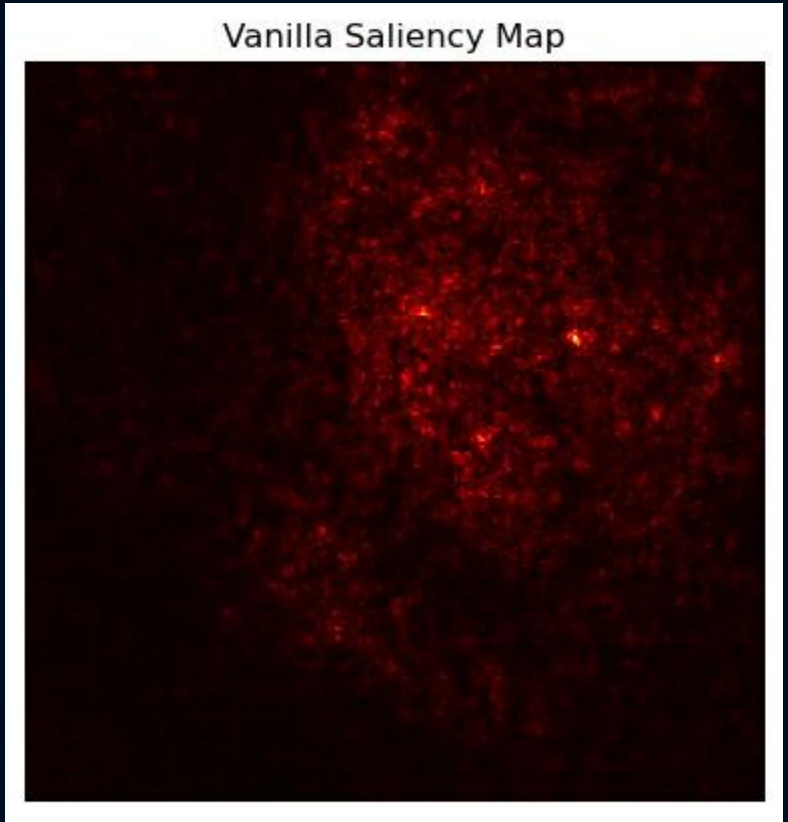
$$M_C(x) = \partial S_C(x) / \partial(x)$$

- Equation represents how much a tiny change in each pixel of the input x affects the final classification score.

- ***Algorithm***
  - A **forward pass** through the network is performed with the **input x** to be able to calculate the **activation function S_C** for **the target class c**.
  - A **backpropagation pass** is performed to be able to **find the gradient of S_c** with respect to the input pixels.
  - Optionally, the **absolute value of the gradients** can be taken or a **limit** can be put on any outlying values to create a more visually coherent heatmap.

VANILLA SALIENCY MAPS

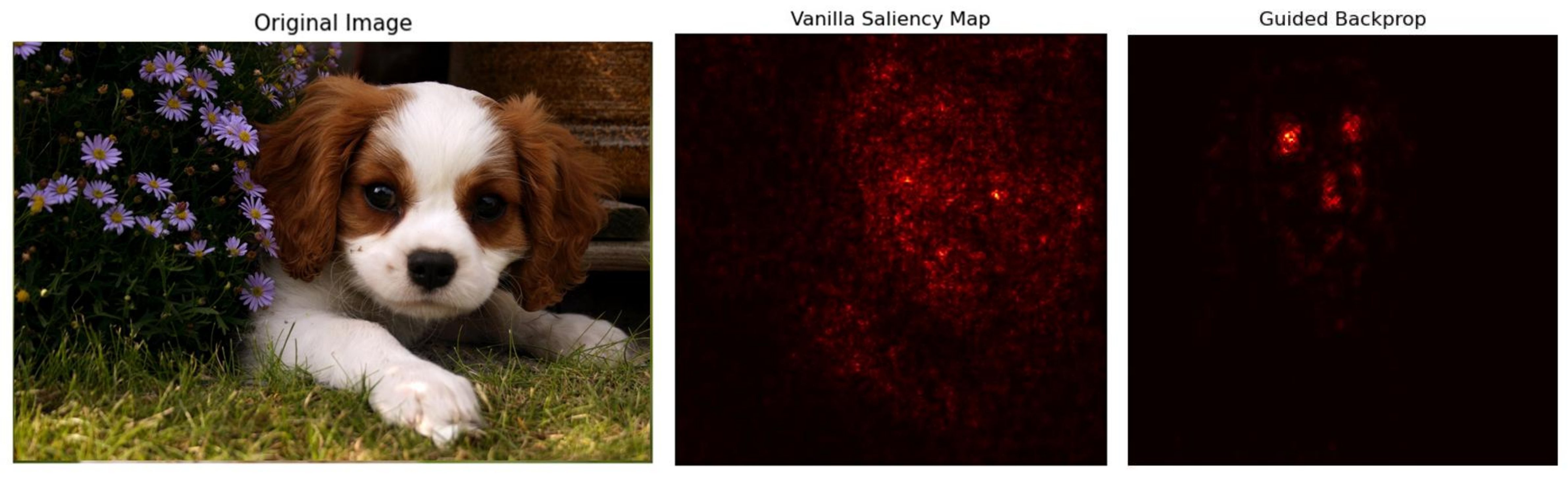
Gradient Calculation



GUIDED BACKPROPAGATION

- This is designed as an **improvement for Vanilla Saliency Map**.
  - To prioritise positive contributions to the final prediction
  - Done by modifying the standard backpropagation algorithm
- Guided backpropagation adjusts the calculation of the input gradients at the **Rectified Linear Unit (ReLU) layers**.
- ***<u>Algorithm</u>***
  - A **standard forward pass** is performed to be able to compute activations.
  - During backpropagation, at each ReLU layer, any **negative gradient values are discarded**.
  - Effectively, the zero-masks gradients where the gradient is negative or where the input activation was negative, focusing the visualization on features that positively triggered high-level units.

GUIDED BACKPROPAGATION



SMOOTHGRAD

- Used to “**sharpen” any noisy gradient-based maps**
  - Done by averaging the gradients of many versions of the same image perturbed with noise

$$\hat{M}_c(x) = 1/n \sum_{i=1}^{n} M_c(x + N(0, \sigma^2))$$

n = the number of noise trials,

N (0, $\boldsymbol{\sigma}^2$) = the Gaussian noise with standard deviation

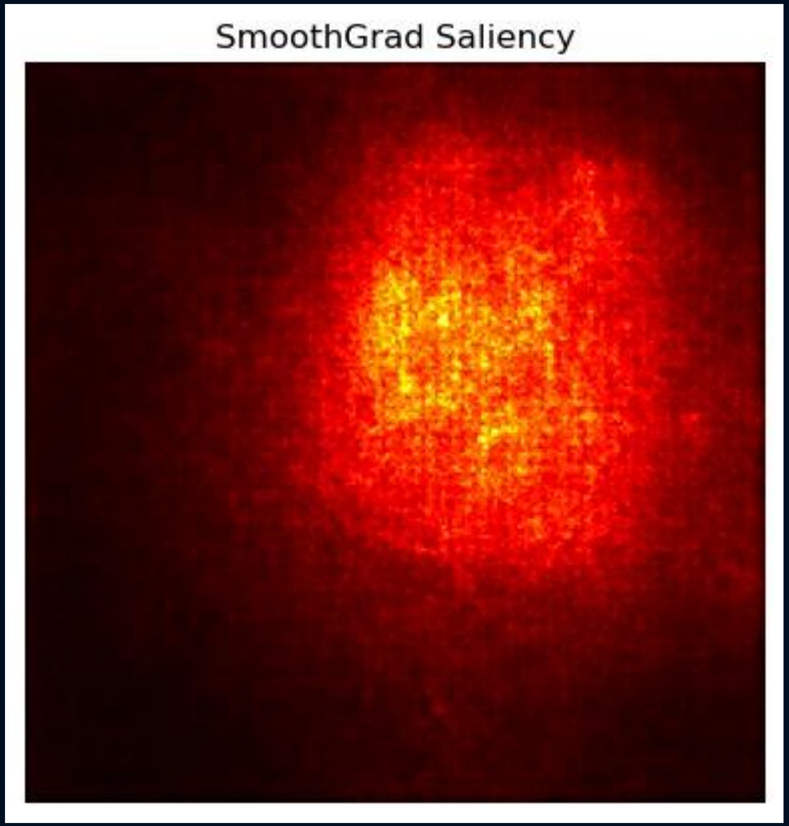
σ $\hat{M}_c$ = the smoothed gradient of M_C (x).

$\hat{M}_c$ = the average of the model’s sensitivity measured according to slightly distorted maps of n noisy samples

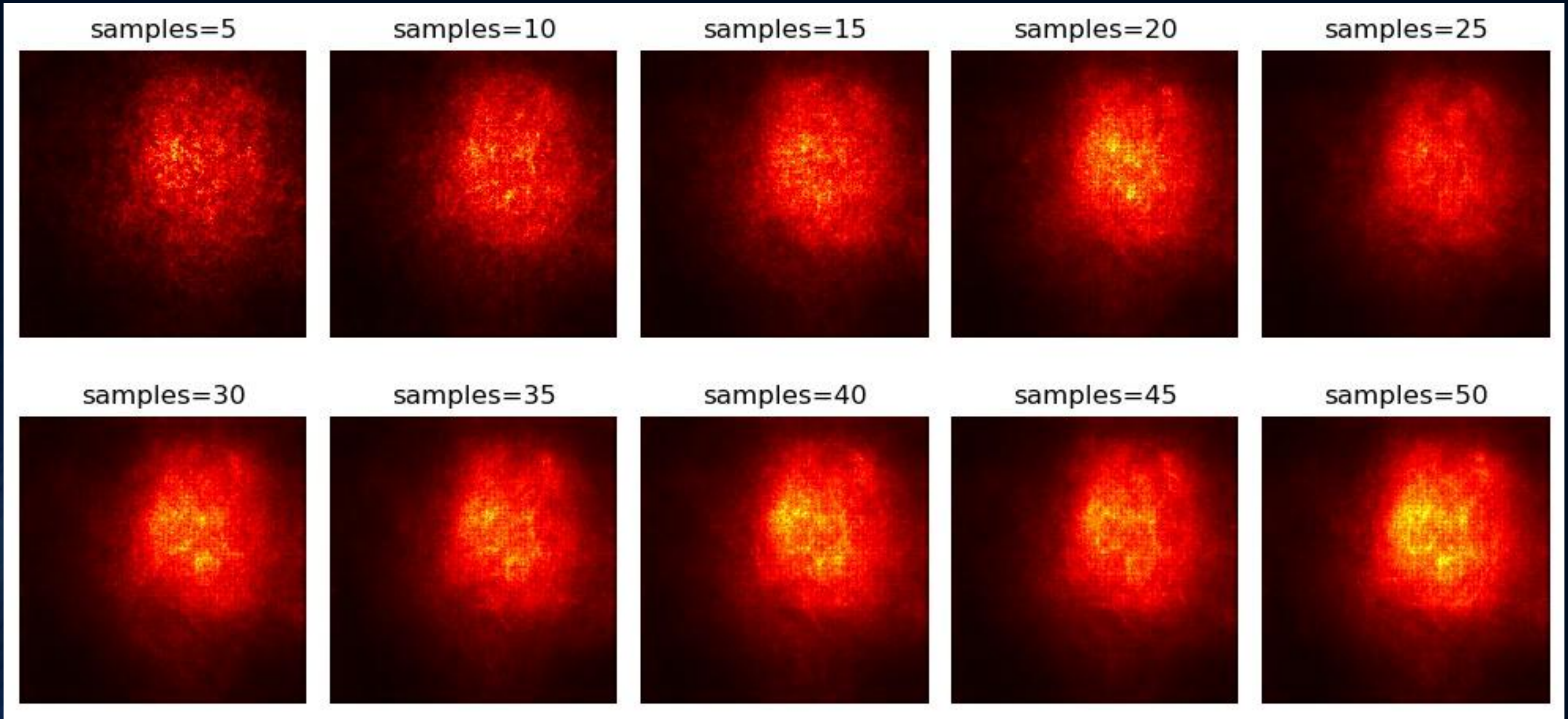
- ***Algorithm***:
  - A sample size **n** and a noise level **σ** are selected.
  - **N** perturbed images are then generated by adding random Gaussian noise to the original input **x**.
  - The vanilla gradient (sensitivity map) for each of the n perturbed images is calculated.
  - The average of these n gradients is then calculated to produce the final sharpened saliency map.

SMOOTHGRAD

Vanilla Saliency
+ Average



SMOOTHGRAD



samples=5
samples=10
samples=15
samples=20
samples=25
samples=30
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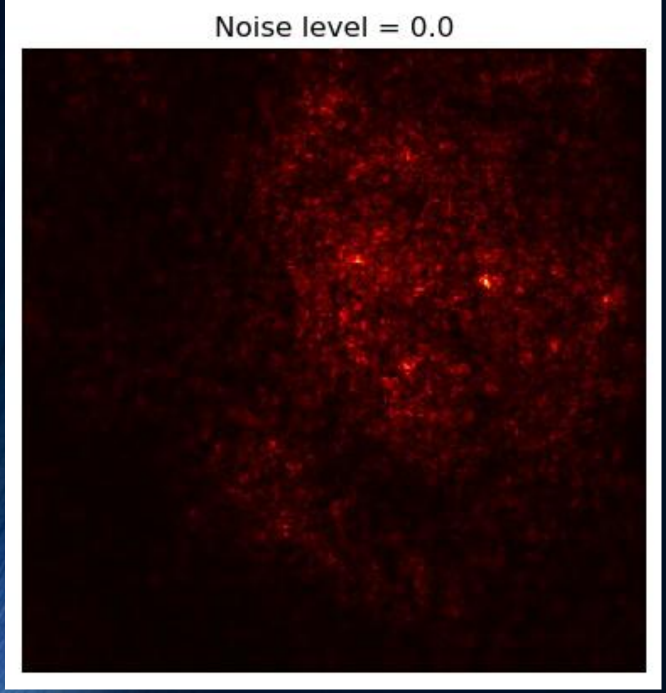
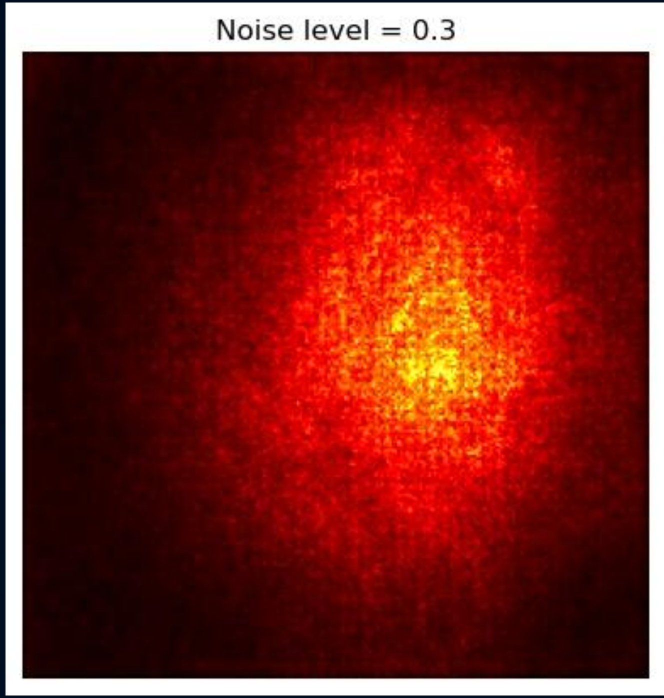
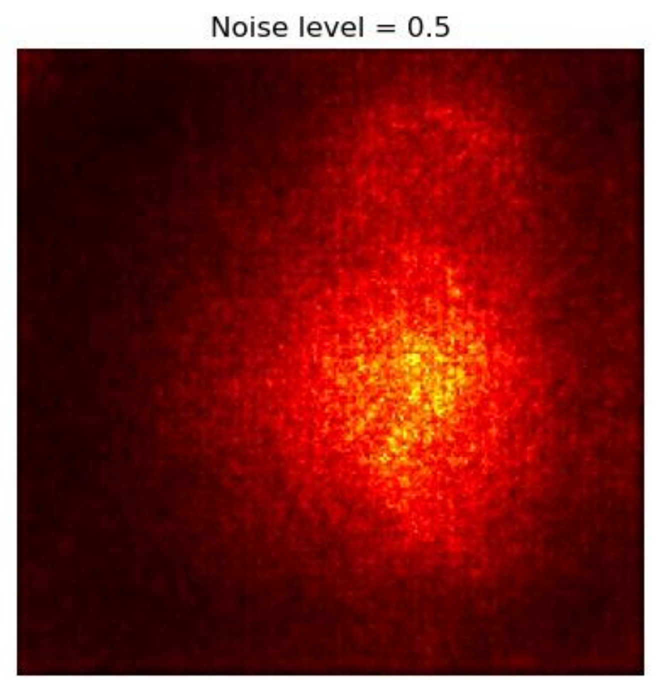
samples=50

SMOOTHGRAD







Streamlit

localhost:8501

UEFA.COM Facebook Inbox (65) - lucadeg... Messenger SchoolHandle - Sch... Instagram CodeSprint Twitch Adobe Acrobat

Deploy

Saliency Map Interactive Simulator

Upload an image

Upload 200MB per file

Overlay Transparency

0.50

Enable Failure Mode (Random Noise Input)

Method A

Vanilla

Method B

Vanilla

Explanation

Vanilla gradients can be noisy because they are highly sensitive to small input changes. Guided Backprop filters negative gradients to produce sharper visualisations. SmoothGrad reduces noise by averaging gradients over multiple noisy inputs.

WHEN SALIENCY MAPS FAIL

- Some saliency methods produce **visualisations** that are **independent of both the model's weights and the data generation process**.

This problem is known as the **Edge Detector Problem**.

- Many saliency methods are unreliable because they **lack input invariance**.
  - Input invariance is a requirement that an attribution method needs to replicate the sensitivity of the model regardless of how the input is transformed.

- Many existing methods fail to satisfy sensitivity and implementation invariance.
- **Raw gradient-based maps** are **visually noisy**, **highlighting pixels** that can appear randomly to the human eye.

WHEN NOT TO USE SALIENCY MAPS

SCENARIO	LIMITATION	BETTER ALTERNATIVE
Strong theoretical guarantees are needed.	With saliency maps, basic gradient methods can violate **sensitivity** and **implementation invariance**.	Provide stronger theoretical guarantees using integrated gradients
Spatial localisation is more important than the sensitivity at pixel-level.	With saliency maps, the gradients at pixel-level can be **noisy** and difficult to interpret.	Use techniques such as **Grad-CAM** to show important areas at a higher semantic level
Explaining **black-box models**	When using saliency maps, any gradient-based saliency requires **differentiable neural networks**.	Model **agnostic explanation methods** such as **LIME** can be used.

REAL WORLD APPLICATIONS

Medical Diagnosis

Did the model focus on pathology?

Autonomous Driving

Did the car stop in front of a fire truck?

Security Systems

What visual evidence triggered it?

Interpretability allows trust, debugging and interpretability.

CONCLUSION

- Saliency maps give us a way to **visually inspect how neural networks make image-based decisions**.
- **Input gradients** form mathematical basis for methods such as Vanilla Saliency, Guided Backpropagation and SmoothGrad.
- These methods help **improve transparency** and **support model debugging** in real-world scenarios.
- Despite their usefulness, saliency maps can lead to explanations that are noisy and misleading.
- They should be used critically and alongside different interpretability methods when an in-depth explanation is required.

THANK YOU FOR YOUR ATTENTION